# Gravitational potential energy



**antocciphysics | TEACHER KEY | CPA**

**Learning target:** Relate height change to near-surface gravitational energy.

**Suggested first pass:** 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

**01. Bring back an earlier idea (retrieve)** *Review: Unit 4, Day 4; 3 course days later.*
$W_{net}$=-12 J; speed falls for fixed mass.

**02. Explain a worked example (worked)**
Δh=3 m, ΔU=60 J; absolute heights may use any common reference.

**03. Look and explain (model)**
Higher book; for equal mass, energy proportional to height.

**04. Build the solution (practice)**
20 J; 60 J; change 40 J.

**05. Use the idea (reason)**
No. It means location below chosen reference; differences matter.

**06. Choose a method (mixed)** *Review: Unit 3, Day 9; 12 course days later.*
They exchange velocities: A=-4 m/s; B=+6 m/s.

**07. Exit check (exit)**
60 J.

**08. STRETCH: Apply it another way (stretch)**
Examples 1 kg raised 6 m and 3 kg raised 2 m; positive change 60 J.

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**01. Bring back an earlier idea (retrieve)** *Review: Unit 4, Day 4; 3 course days later.*
$W_{net}=\frac{1}{2}m(v_f^2-v_i^2)=-36$ J, negative as kinetic energy falls.

**02. Explain a worked example (worked)**
$\Delta h=3$ m, $\Delta U=60$ J; absolute heights may use any common reference.

**03. Represent and interpret (model)**
Higher book; for equal mass, energy proportional to height.

**04. Solve and justify (practice)**
$\Delta U$ 250 J either way; absolute U changes, difference does not.

**05. Analyze the reasoning (reason)**
No. It means location below chosen reference; differences matter.

**06. Choose a method (mixed)** *Review: Unit 3, Day 9; 12 course days later.*
$v_f=1$ m/s; $K_i=104$ J, $K_f=4$ J. K decreases by 100 J; perfectly inelastic.

**07. Exit check (exit)**
$\Delta U=-80$ J; gravity work +80 J.

**08. STRETCH: Apply it another way (stretch)**
$U_i=-60$ J, $U_f=150$ J; $\Delta U=210$ J; gravity does -210 J.